BACSE324 · EMBEDDED SYSTEMS ARCHITECTURE · FALL 2026 · REVIEW 2

Design Models: CDFG, FSM and Task Schedule

Review 2 submission · due end of September · 6–10 pages excluding cover and appendices

Project title	
Scope tier (from review sheet)	
Submission date	
Repository link (optional)	

TEAM

Member	Reg. no.	Module owned	Sections written

Grey shaded text is guidance. Delete every guidance block before you submit. Keep the section numbers and headings — markers read every report in the same order.

Diagrams must be your own — draw.io, yEd, Visio or a clean hand drawing scanned at legible resolution. A screenshot of a tool-generated or AI-generated diagram you cannot explain scores band 1. Every number needs a unit and a source: measured, datasheet or estimated.

1 System recap

Half a page. One paragraph on what the device does and for whom, then a block diagram. This is context only; it is not marked on its own.

Insert figure here — replace this box

Figure 1 — System block diagram: MCU, sensors, actuators, communication links, power.

1.1 Conditions carried from proposal review

Copy each condition of approval and each Review 2 mandatory addition from your review sheet. These are ticked off on your marking sheet (4 marks).

Condition / mandatory addition (from your review sheet)	Status	Where it is addressed

2 Control Data Flow Graph

Choose one function, ISR or processing loop — the one you believe is the critical path of your system. You are marked on the justification of that choice as much as on the graph.

2.1 Choosing the critical function

List at least three candidate functions or ISRs. Say why the chosen one is critical: it has the tightest deadline, the longest execution, the most data dependencies, or it blocks other work.

Function / ISR	Trigger	Rate	Est. execution time	Deadline	Chosen? Why / why not

Justification:

2.2 The CDFG

Notation: operation nodes as circles, decision nodes as diamonds, data edges solid, control edges dashed. Group nodes into basic blocks. Label each node with its operation and its estimated cost in cycles or µs.

Insert figure here — replace this box

Figure 2 — CDFG of [function name].

2.3 Node table

Node	Operation	Inputs	Output	Est. cost (cycles / µs)	Source line or function

2.4 Critical path

List the nodes on the longest path, total their cost, and compare it with the deadline from 2.1. Then name one dependency that could be broken, reordered or run in parallel — and whether you will.

Critical path (node sequence)	Total latency	Deadline	Slack

What the CDFG tells us about the implementation:

3 System-behaviour FSM

Model the operating modes of the whole device — idle, armed, measuring, transmitting, fault and so on — not the internals of one function.

3.1 Model type

State whether this is a Moore machine (outputs depend on state only) or a Mealy machine (outputs depend on state and input), and why that fits your device.

Model type: Moore / Mealy —

3.2 State diagram

Insert figure here — replace this box

Figure 3 — System FSM. Label every transition as event [guard] / action.

3.3 State transition table

Current state	Event	Guard	Next state	Output / action

3.4 Fault, recovery and safe state

At least one fault state is required. Define the safe state: what the outputs are when the device is in it, and how it is reached from every other state.

Safe state and outputs:

3.5 Completeness check

For each state, every event must either cause a transition or be explicitly ignored. Fill the matrix with the next state, or a dash for "ignored".

State ↓ / Event →				

4 Task schedule

4.1 Task set

Every periodic task, ISR and event-driven job. Say how each WCET was obtained: measured (GPIO toggle with a scope or logic analyser, or a cycle counter), from the datasheet, or estimated from the CDFG.

Task	Trigger	Period T	WCET C	Deadline D	Priority	U = C / T	WCET source

Total utilisation U =

4.2 Scheduling approach

Choose one approach and justify it. Task level: cyclic executive, rate-monotonic (RMS) or earliest-deadline-first (EDF). Operation level, applied to your CDFG: ASAP, ALAP, mobility analysis, resource-constrained list scheduling, force-directed or ILP.

Approach chosen:

Why this approach and not the obvious alternative:

4.3 Schedulability analysis

Show the working that proves the schedule meets its deadlines. RMS: utilisation bound U ≤ n(2^(1/n) - 1), or response-time analysis if the bound fails. EDF: U ≤ 1. Cyclic executive: major and minor cycle, and the frame table. ASAP / ALAP / list: the control-step table with mobility per node.

4.4 Timeline

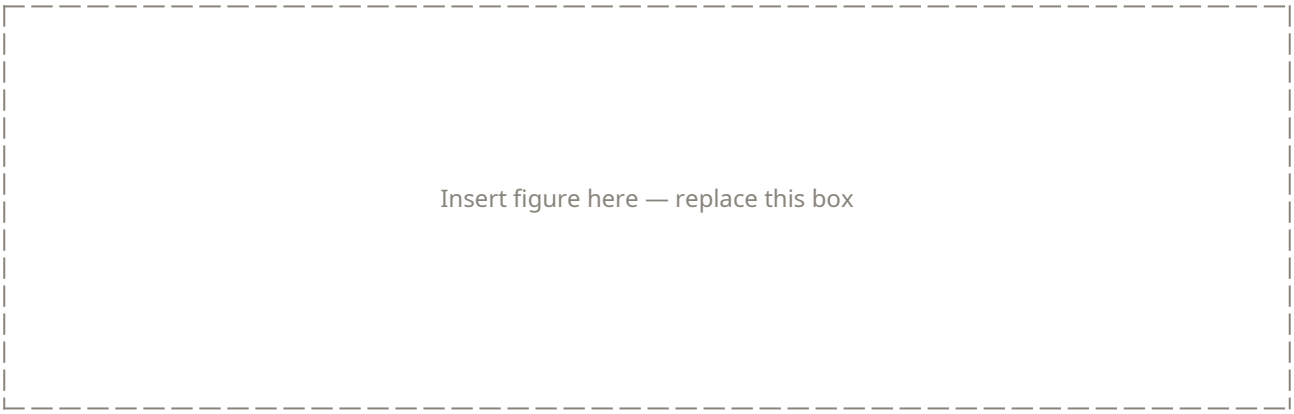


Figure 4 — Schedule over one hyperperiod (task level) or across all control steps (operation level).

4.5 Shared resources and interrupts

What data or peripherals are shared between tasks or with ISRs, how they are protected (critical section, flag, queue, double buffer), and the worst-case blocking that adds.

Shared resource	Used by	Protection	Worst-case blocking

5 Planned traceability

Map each model element to the source file and function you will write. The Final report extends this table to match the code as built.

Model element (CDFG node, FSM state, task)	Planned file	Planned function	Owner

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6 Plan to Final

Remaining work	Owner	Target date	Risk and fallback

7 Individual contribution

Who produced which model. Each member must be able to explain the model they drew at the review.

Member	Model / section produced	Module for Final	Share %

References

IEEE style. Cite datasheets for every timing figure you did not measure.

[1]

Submission checklist

Item	✓
Guidance blocks deleted; 6–10 pages excluding cover and appendices	
Critical function justified against at least three candidates	
CDFG with node costs and the critical path stated	
FSM with model type, transition table, fault and safe state	
Task set with WCET source, chosen approach and schedule-ability proof	

Item	✔
Every carried condition listed with its status	
Every member named against a model they can explain	